2022 International Conference on Emerging Smart Computing and Informatics (ESCI)
AISSMS Institute of Information Technology, Pune, India. Mar 9-11, 2022

Education and Analysis of Autistic Patients Using Machine Learning

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2022 International Conference on Emerging Smart Computing and Informatics (ESCI) | 978-1-6654-0073-2/22/$31.00 ©2022 IEEE | DOI: 10.1109/ESCI53509.2022.9758322

***Abstract*— "Autism is a neurological condition that decapitates patient of development in language and communication skills restraining him/her from any social interaction and develops rigid, ardent behavior." To overcome this problem we have come up with an educational app that helps people to get educated via friendly UI while under surveillance of personal or assigned therapist. Features include fast and effective teaching in vocabulary, therapist can personalize their education methods to make model effective, analysis system that will track every move of patient and utilize local medical AI records to report patient's metric progress and pattern of impairment triggers. Audio to Visual and vice-versa. Pronounce words and check whether input correct or not. The app is open source mainly to educate the people who are suffering from autism. It will benefit patient to gain knowledge of communication. Also, they would get something that will accompany them all time, so that they won't feel lonely also ensuring anxiety risk in case of physical therapy methods. Requirements of the application is implemented by using Java Struts, JDBC, XHTML, CSS, JavaScript, and DOM.**

Keywords—Machine learning, Software, Autism Disorder, Educational therapy.

I. INTRODUCTION

A. Problem Statement

Patients with ASD faces high mental issues and challenges related to senses required in social, academics and communication. There's special requirement of education method of these patients involving separate training in academics and social skills. Proposed education method using interactive graphical makes it easy for students to learn where inputs from student arranges visual manner of study.

B. Problem Solution

Machine learning model to identify abnormal function of resting state network in Autism combined with therapist interaction to develop a system that will help in education of patient without any physical involvement. Support vector machine and independent component analysis over image based meta-analysis. We are utilizing biomarker of ASD patients available publicly at ABIDE dataset consisting of more than 432 patients and matched with 556 healthy controls. Upon pre-processing model will detect pattern of outbursts which should not be tampered with. Then model will apply emphatical values in education process with clouded involvement of therapist to teach the patient through interactive touch and speak based application. In total application will utilize machine learning to detect human behavior and then suggest training methods for recovering the weak spots.

II. LITERATURE SURVEY

A. Background Information

Autism medical term was introduced back in 1911 by Bleuler. His report explained effects on behavior of children struck by autism. It was found that there's similarity in pattern of each patient that now is making it helpful for treatment of such.

The cause of autism is still a question to medical researchers however there's a breakthrough that this disorder can be inherited by family over decades, numerous studies show subjects were decapacitated of basic motor and logical functions.[1]

B. How Autism affect people

There are series of effects on patients with this disorder:

- Autistic patients tend to lack in communication skills and struggle to interact.
- Patients are sensitive to sound or touch or light colors.
- Patients lack emotion and logical key points which differ them from a normal healthy lifestyle.
- Autism can strike due to high intensity trauma to brain making them fear anything related to that case.
- Even though lacking many healthy requirements study shows autistic patients have good remembering capabilities and can grasp knowledge on a particular subject.

C. Statistical Diagnosis on publication and awarness spreading

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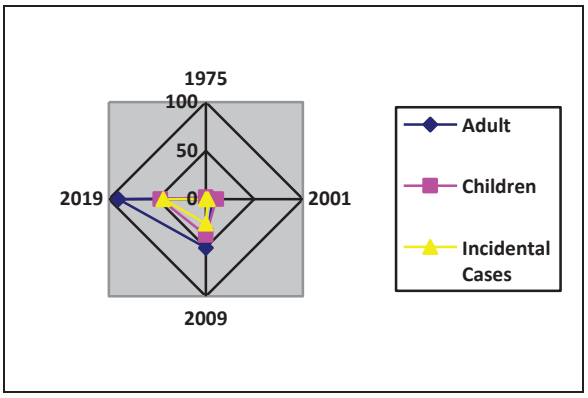


Fig. 1. Graph depicting diagnosis of ASD in individuals over time

With time research on this medical condition took a pace and many facilities were set up for this agenda to diagnose and treat the patients with Autism. General awareness camps are being setup and medical staff are being trained for coping with specially aided patients.

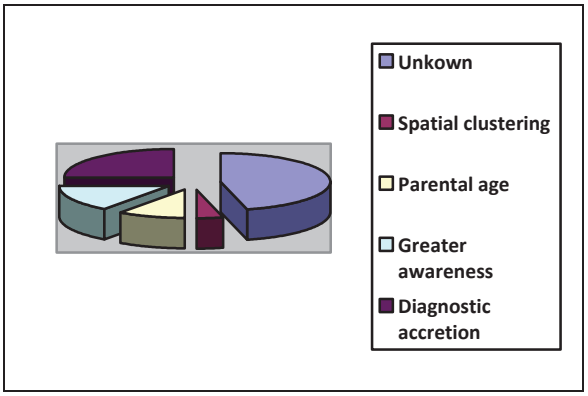


Fig. 2. Reasons of increased diagnosis of autism

III. DATASET

ABIDE data is being utilized for this research, it is a pre-processed repository open sourced [3], [4], [5]. Research includes 3 pipeline version. Detailed summary of datasets is given below:

TABLE I. PIPELINE VERSION

Sr. No.	Dataset type	Sources	Attribute Type	Number of attributes	Number of instances
1	CCS	ABIDE Repository	Boundary-based rigid body (BBR)	19	1002
2	NIAK	ABIDE Repository	Objective Dual Regression	19	232
3	DPARSF_fMRI	ABIDE Repository	Categorical, continuous and binary	19	187

Below are the attributes based on which prediction is optimized

Attribute Id	Attributes Description
1	Patient age
2	Sex
3	Nationality
4	Jaundice Case: Y/N
5	Screening method
6-15	10 question quiz
16	Screening Score

TABLE II. ATTRIBUTES IN DATABASE

IV. METHOD TO IMPLEMENT

Figure 1, 2 and 3 depicts process involved in our proposal is further explained in detail below.

ASD Raw Dataset (input) Pre-processed data Run ML algorithm: CNN , SVM and LR Data processing: Normalization Classification of ASD and Non-ASD traits(Output)

Figure 1. Autism Classification

User Inputs Apply ML algorithm LR and NB Comparison of user to open data Fetch Training data

Figure 2. User to Data comparison

A. Data Pre-Processing

Data pre-processing is method required to transform raw data provided mostly open source to information that can be used in proper format. Yield of human submitted data is compartively low due to presence of errors and null values. Pre-processesing method gives better yield as it removes inconsistency of data and missing dta is omitteed using imputation method.

B. Training data

The dataset needs to be split into two parts, for training and testing purpose. Traditionally ratio is of 80:20 for training dataset and testing dataset respectively.

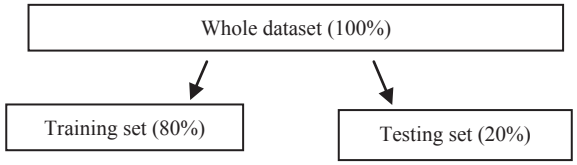


Fig. 3. Training and Testing set

C. Machine learning

1) Convolutional Neural Network: CNN is learning technique for constructing problem models.[7].
Involves many layers ranging from input to output with normalization, matrix operation, pooling layers.

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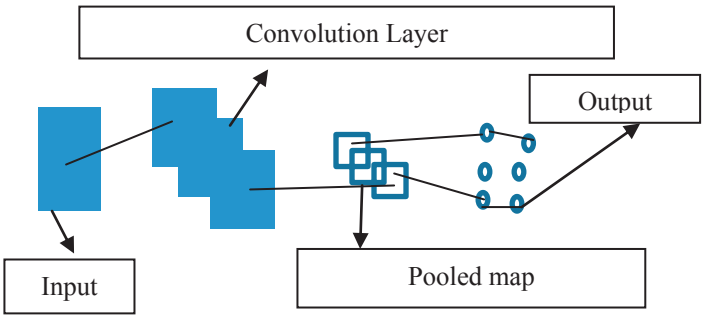


Fig. 4. CNN Model

2) Logistic Regression (LR):

It is regression for analysing binary depended variables. Outputs are termed as 0 and 1. It is also represented as sigmoidal function.

3) Support Vector Machine):

It is supervised learning methoda required for various classification and regression by recognizing patterns in data and re defining boundaries. [6]

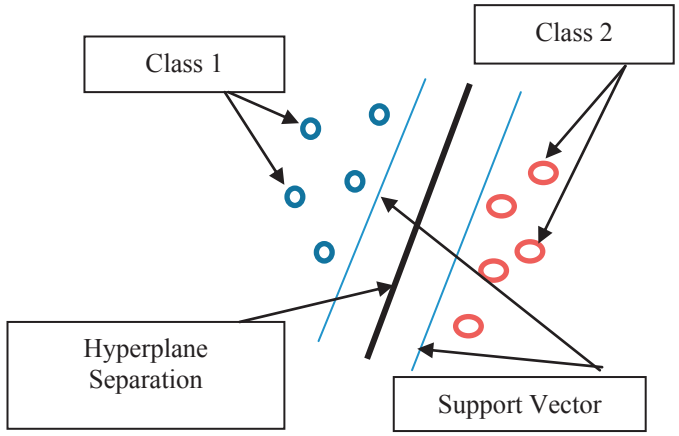


Fig. 5. SVM Classifier

D. Model for detection

1) Data exploration

Raw data set contains 1054 with 109 individuals diagnosed with ASD. So diagnosed rate is 10.341

Sample as shown below.

	Case_No	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10	Age_Mons	Qchat-10-Score	Sex	Ethnicity	Jaundice	Family_mem_with_ASD	Who completed the test	Class/ASD Traits
0	1	0	0	0	0	0	0	1	1	0	1	28	3	f	middle eastern	yes	no	family member	No
1	2	1	1	0	0	0	1	1	0	0	0	36	4	m	White European	yes	no	family member	Yes
2	3	1	0	0	0	0	0	1	1	0	1	36	4	m	middle eastern	yes	no	family member	Yes
3	4	1	1	1	1	1	1	1	1	1	1	24	10	m	Hispanic	no	no	family member	Yes
4	5	1	1	0	1	1	1	1	1	1	1	20	9	f	White European	no	yes	family member	Yes

Fig. 6. Dataset

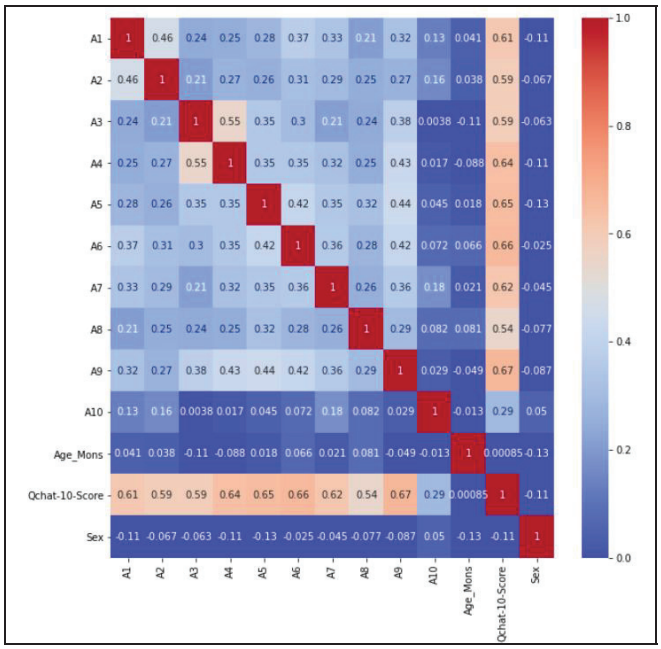


Fig. 7. ASD Statistics

2) Visualization

Axes plot which represents distribution of ASD diagnosis. Blue represents non-autistics while orange is for autistic patients.

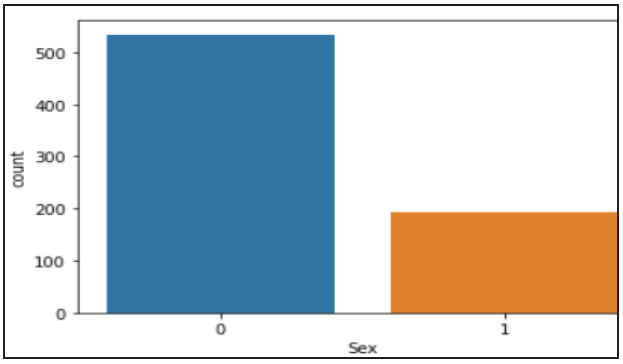


Fig. 8. Fact plot: 'Axes'

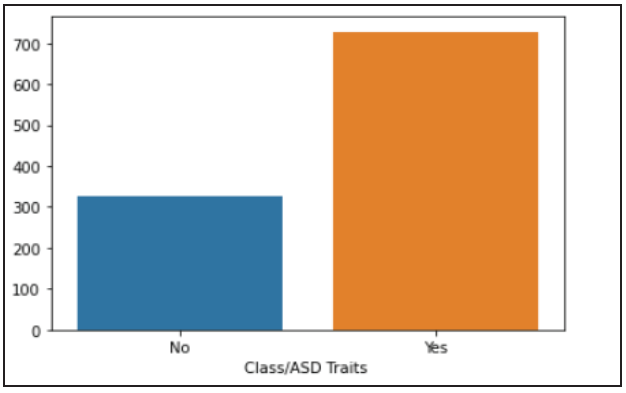


Fig. 9. Fact plot: 'Axes'

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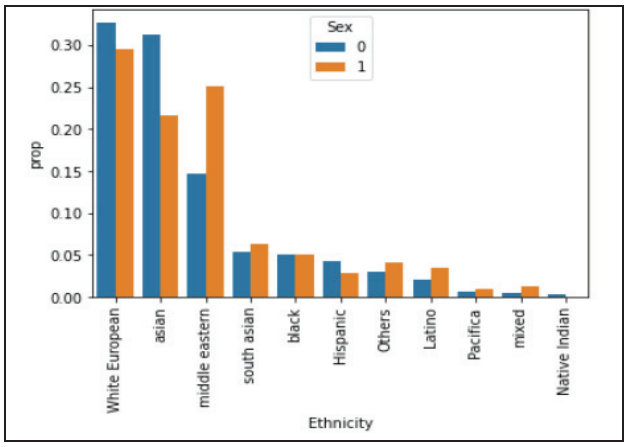


Fig. 10. Fact plot: 'Axes'

Next Figure. 9 is fact plot displaying result score respective to age in months.

Blue dots r

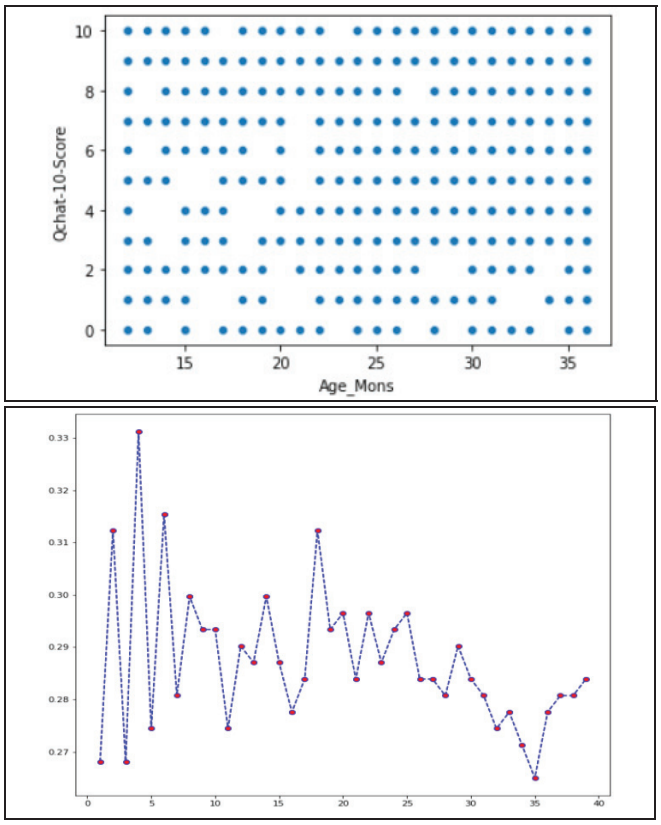


Fig. 11. Fact plot: 'swamp'

All blue series shows detection of data response to test. Where result is high and very likely to relate to autism class.

E. Method of teaching

It is observed that patients if ASD shows higher capacity of visual and photographic strength. In order to assess intellectual part, we aim to teach them using graphical learning. This application will provide remote education where therapist and our machine learning will work together to find better way in education domain. Concept is to analyze trigger point and weak spots in development of child and to overcome that area.[2]

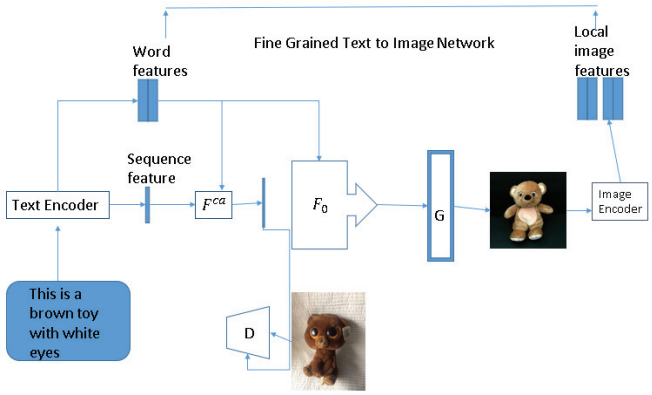


Fig. 12. Neural network for text-image[8]

V. RESULT AND DISCUSSION

A. Evaluation

Classification model needs to be gone through extreme scrutiny to work and achieve its goal. Effectiveness performance metric is subject to contributor because logical and mathematical representation of submitted model varies from mind to mind.

TABLE III. ELEMENT OF PREDICTIVE MATRIX

	Predictive ASD values	
Actual ASD values	TP	FP
	FN	TN

$$\text{Sense Rate} = \frac{TTTT}{[TTTT+TTTT]}$$

$$\text{Sensitivity} = \frac{TTTT}{(TTTT+FFTT)}$$

$$\text{Accuracy} = \frac{TTTT+TTTT}{[TTTT+TTTT+FFTT+FFTT]}$$

Experimental result of ASD screening for adult, children and adolescents.

In CNN learning rate is 001 and 100 for epoch with binary cross loss function 16 and 32 filters of 0.5 dropouts been used.

For SVM 0.1 gamma value set as default.

Results of of these model on Autism Syndrome Disease:

TABLE IV. DATA SCREENED FOR ADULTS

Algorithm	Specificity	Sensitivity	Accuracy (%)
LR	0.95	0.9696	96.69
SVM	0.95	0.88888	98.11
CNN	1.0	0.9939	99.53

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TABLE V. DATA SCREENED FOR CHILDREN

Algorithm	Specificity	Sensitivity	Accuracy (%)
LR	1.0	0.9677	97.40
SVM	1.0	0.9679	96.75
CNN	1.0	0.9678	98.30

All learning curves of machine learning algorithm described below for adults and children.

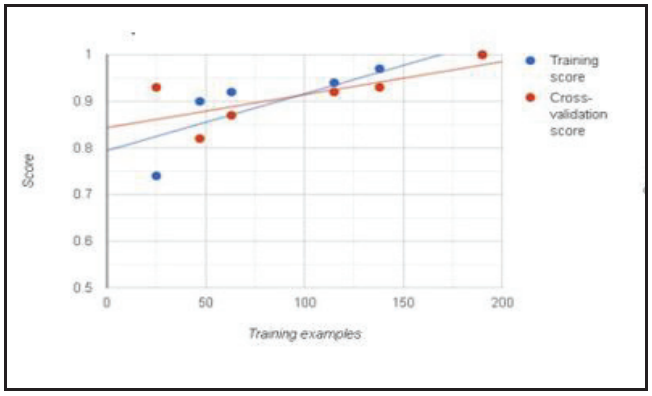


Fig. 13. Graph depicting learning of SVM [Adults]

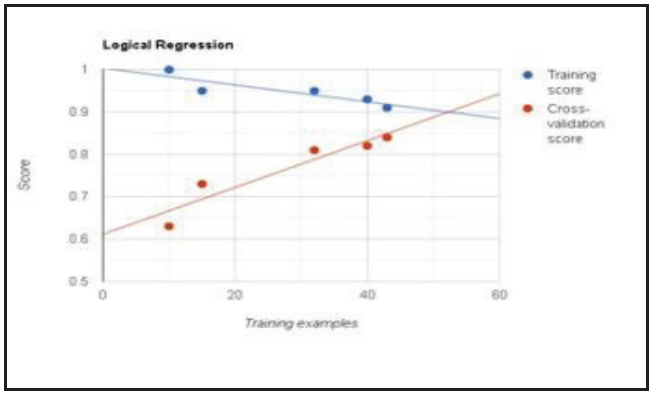


Fig. 14. Graph depicting learning of Logical regression [Adults]

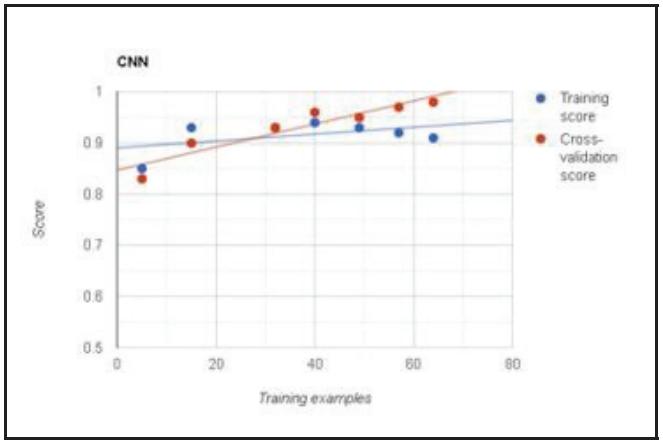


Fig. 15. Graph depicting learning CNN model [Adults]

CNN produced highest 99.53% accuracy for dataset for Adults

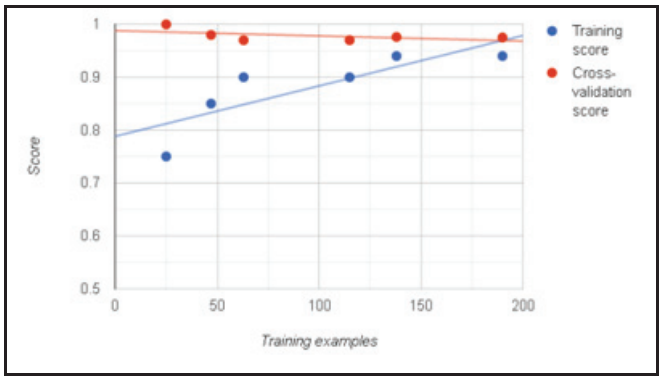


Fig. 16. Graph depicting learning of SVM [Children]

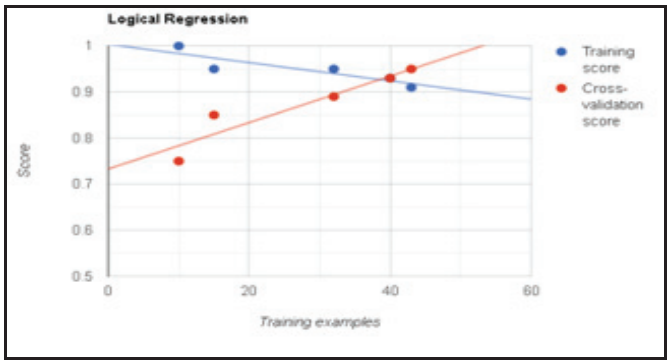


Fig. 17. Graph depicting learning of Logical regression [Children]

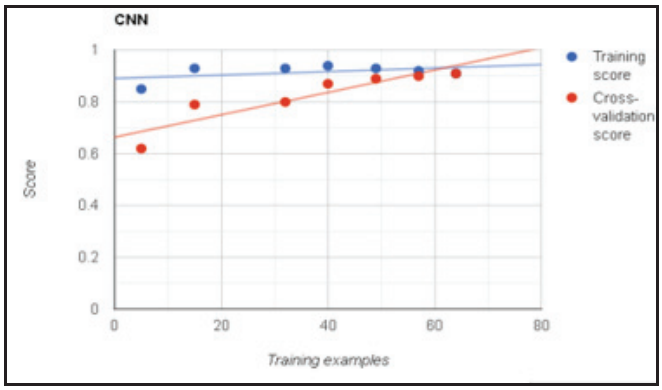


Fig. 18. Graph depicting learning CNN model [Children]

CNN produced highest 98.30% accuracy for dataset for children

VI. CONCLUSION

In this work, detection of ASD and method of education of patient is being attempted using various machine learning techniques. On case study over CNN, SVM and LR we found out CNN accuracy of prediction and suggestion on protocol dominated over other two. CNN model with accuracy of 98-99% can be implemented on model 1 and model 2 so patient can be diagnosed and non-clinical education procedure may start up under supervision of therapist if required.

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